

Processing and interpretation of neuroscience data:

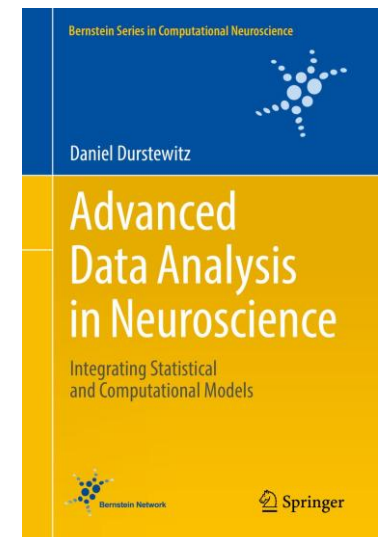
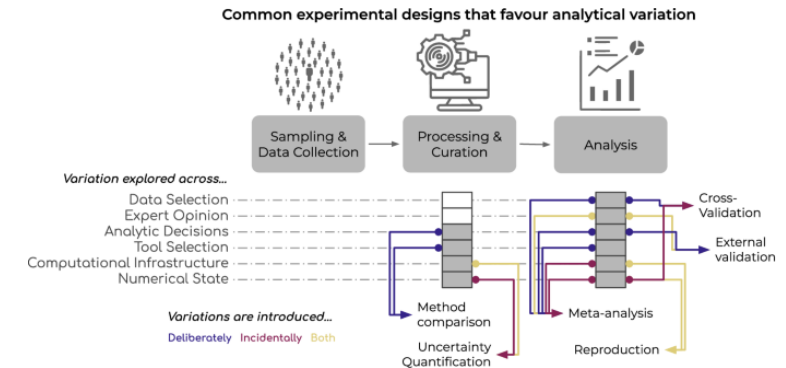
Module 1 – Data overview and single cell sequencing

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Center For Brain Research (Zentrum für Hirnforschung)



Overview

Resources needed:

- Laptop
- Internet access
- Materials:
 - VM machine - Docker
 - <https://github.com/HugoMalagon/NeuroData>
[860.053-MUW](#)

07.10

- Introduction to R, basics
- Visual analytics

14.10

- dimensional reduction: PCA, UMAP
- Normalization/scaling
- clustering: k-means, knn

21.10

- Intro to Seurat
- Single cell RNA seq
- Dataset merging and preprocessing

04.11

- clustering in Seurat
- Finding cell types

11.11

- Joining experiments
- DEG interpretation

„Exam“: Assessment via email after each class

Day 3

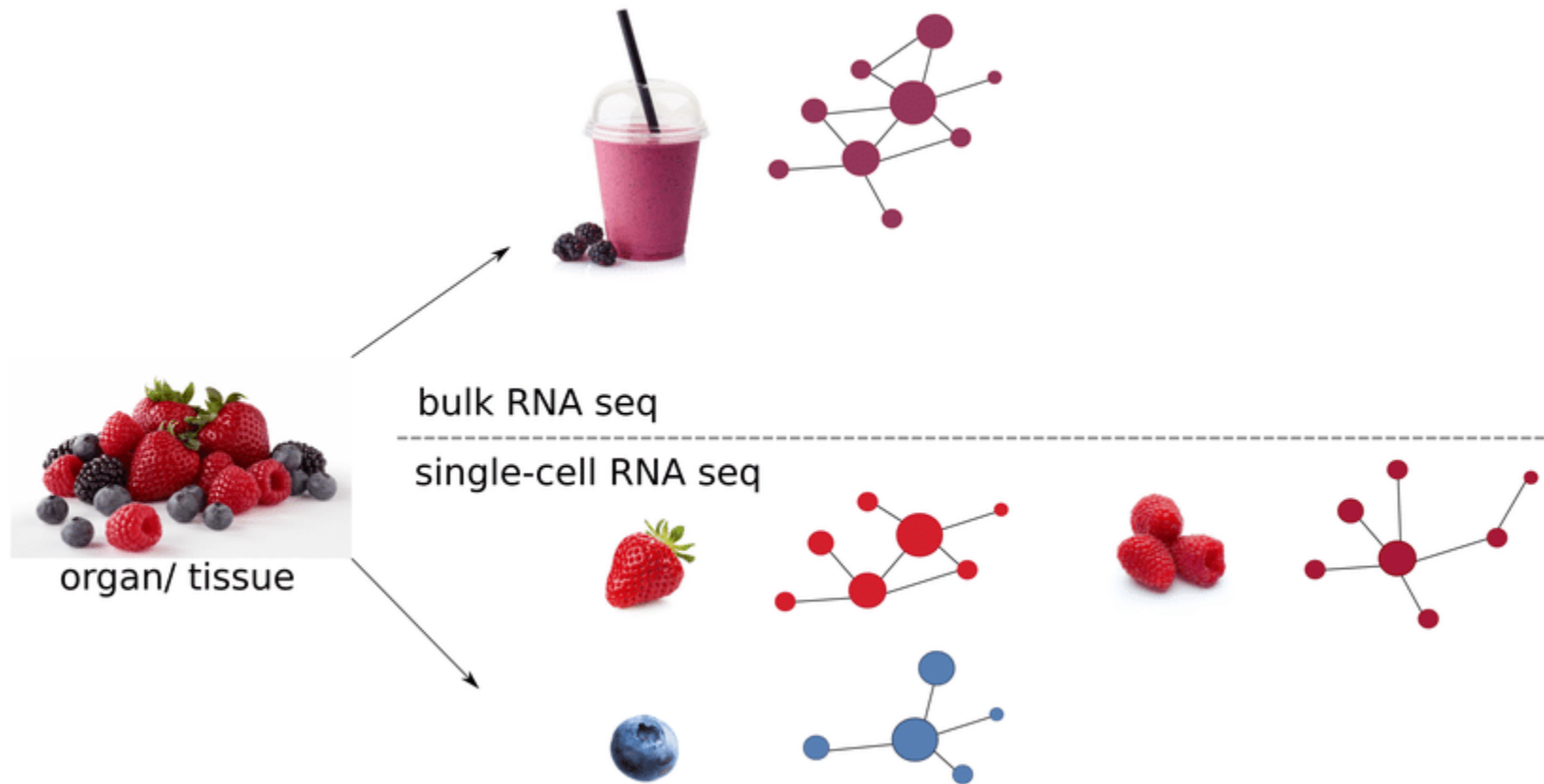
Intro to Seurat

Preprocessing of scRNAseq

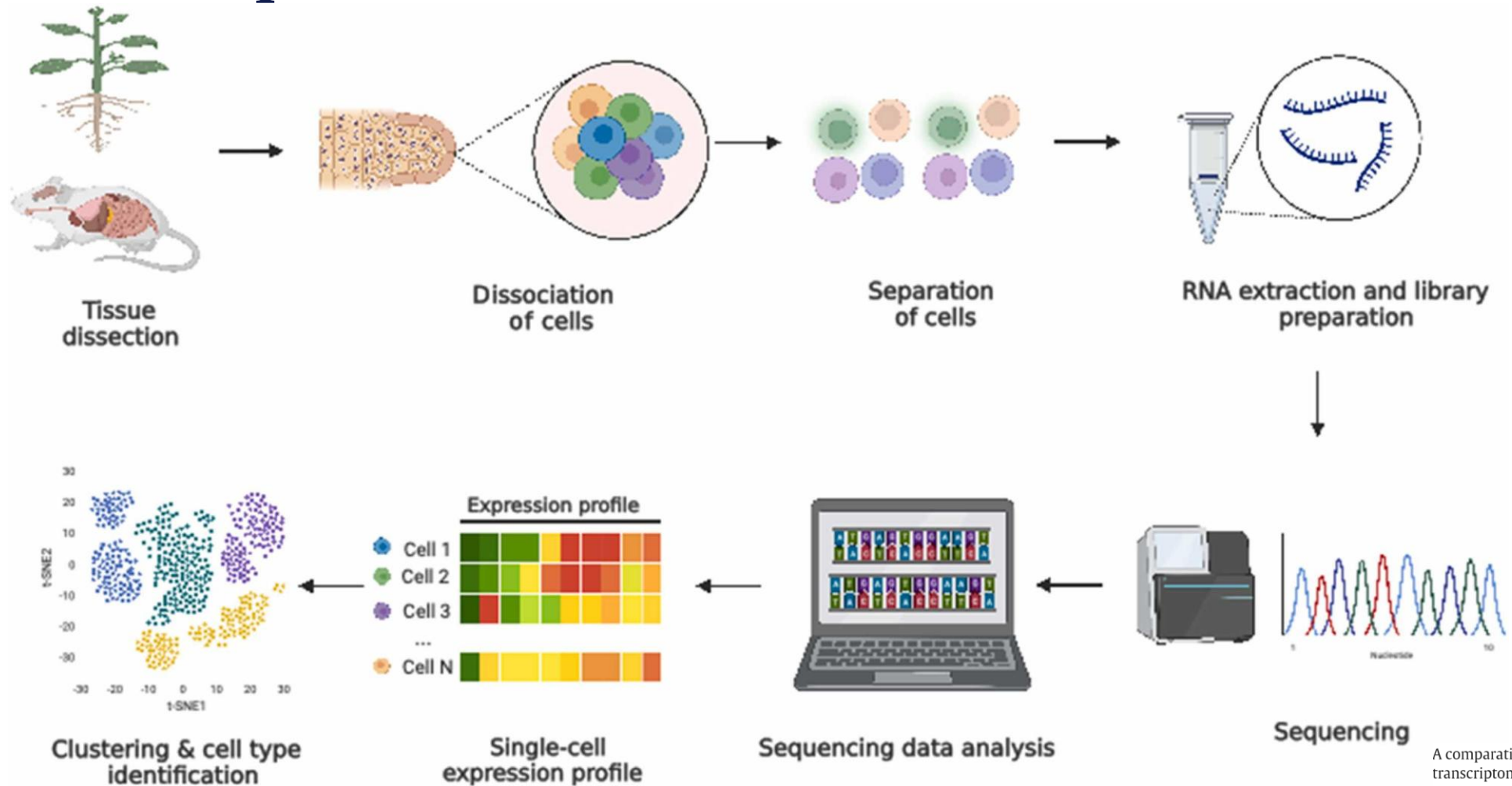
How to measure gene expression?

Method	Throughput	Quantitative	Sensitivity	Main Limitation
Northern Blotting	Low	Semi-quant	Moderate	Low throughput, laborious
qPCR	Low-Medium	Quantitative	High	Target-specific, limited multiplexing
Microarray	High	Quantitative	Moderate	Lower sensitivity than RNA-Seq
RNA-Seq	Very High	Quantitative	Very High	Expensive, computational load
In Situ Hybridization	Low	Qualitative	Moderate	Labor-intensive
SAGE	High	Quantitative	High	Largely replaced by RNA-Seq
Reporter Assays	Variable	Quantitative	Variable	Requires reporter construct
Digital PCR (dPCR)	Low	Quantitative	Very High	Specialized equipment

How to measure gene expression?



Single cell experiments workflow



A comparative analysis of single-cell transcriptomic technologies in plants and animals

Vamsidhar Reddy Netti^{1,2}, Harshraj Shinde^{1,2}, Gulshan Kumar^{1,2}, Ambika Dudhate^{1,2}, Jong Chan Hong^{1,2}, A. B. Uthas Sapanrao Kadam^{1,2}

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<https://doi.org/10.1016/j.cpb.2021.100289>

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Examples of available data sets

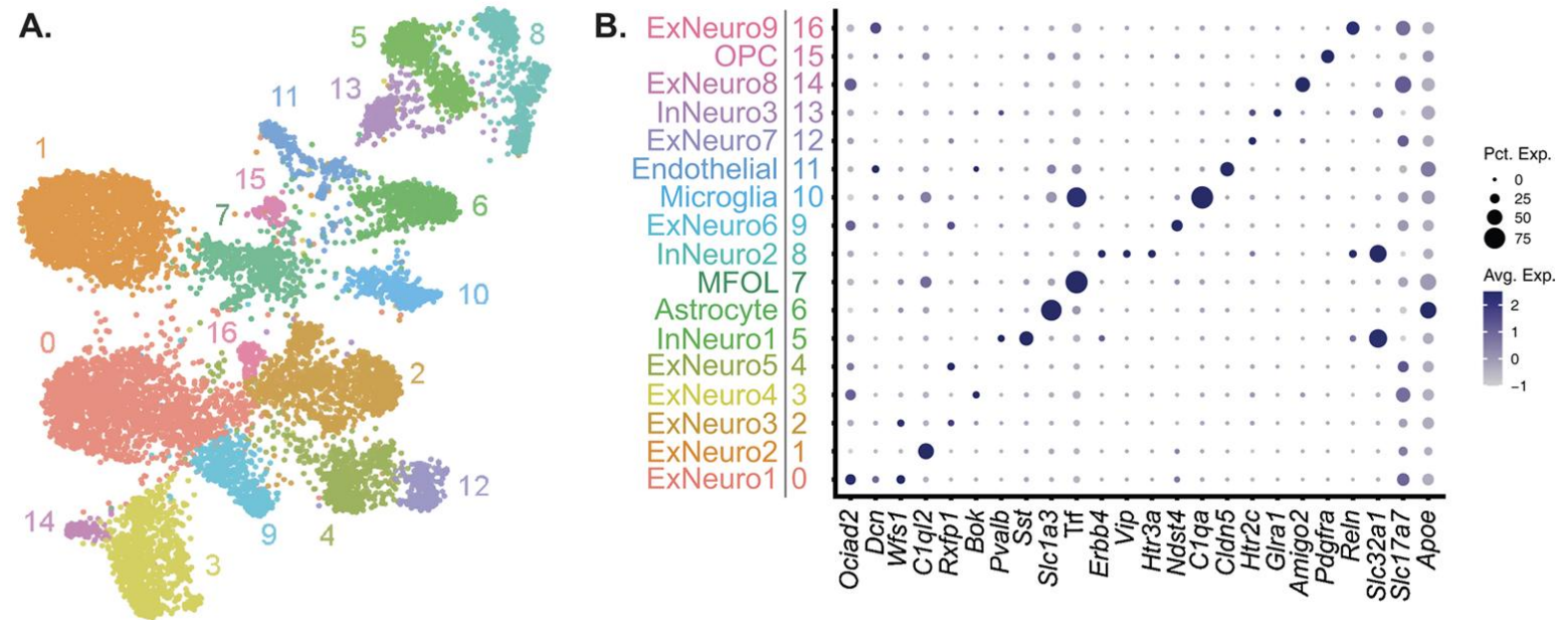
Datasets (Showing 58 datasets)	Product	Species	Sample type	Cells or nuclei	Preservation
320k scFFPE From 8 Human Tissues 320k, 16-Plex	Flex Gene Expression v1.0	 Human	Brain, Colon, Lung, Breast, Lymph node, Kidney, Skin, Cervix	N/A	Fixed
10k Mouse E18 Combined Cortex, Hippocampus and Subventricular Zone Cells, Chromium NextGEM Single Cell 3'	Universal 3' Gene Expression v3.1	 Mouse	Brain	Cells	NA
10k Mouse E18 Combined Cortex, Hippocampus and Subventricular Zone Cells, Chromium GEM-X Single Cell 3'	Universal 3' Gene Expression v4	 Mouse	Brain	Cells	NA
10k Adult Mouse Brain Nuclei Isolated with Chromium Nuclei Isolation Kit, Chromium NextGEM Single Cell 3'	Universal 3' Gene Expression v3.1	 Mouse	Brain	Nuclei	Fresh Frozen
10k Adult Mouse Brain Nuclei Isolated with Chromium Nuclei Isolation Kit, Chromium GEM-X Single Cell 3'	Universal 3' Gene Expression v4	 Mouse	Brain	Nuclei	Fresh Frozen
10k Mouse Forebrain FFPE Tissue Dissociated using gentleMACS Dissociator, Singleplex Sample (Next GEM)	Flex Gene Expression v1.0	 Mouse	brain	Cells	FFPE
Mouse Brain Nuclei Isolated with Chromium Nuclei Isolation Kit, SaltyEZ Protocol, and 10x Complex Tissue DP (CT Sorted and CT Unsorted)	Epi Multiome ATAC + Gene Expression v1.0	 Mouse	brain	Nuclei	Fresh Frozen
Mixture of Lung Cancer and Glioblastoma FFPE Tissues Dissociated Manually or using gentleMACS Dissociator, Multiplexed Samples, 4 Probe Barcodes (Next GEM)	Flex Gene Expression v1.0	 Human	lung, brain	Cells	FFPE
5k Adult Mouse Brain Nuclei Isolated with Chromium Nuclei Isolation Kit	Universal 3' Gene Expression v3.1	 Mouse	brain	Nuclei	Fresh Frozen
8k Adult Mouse Cortex Cells, ATAC v2, Chromium Controller	Epi ATAC v2	 Mouse	brain	Nuclei	N/A
8k Adult Mouse Cortex Cells, ATAC v2, Chromium X	Epi ATAC v2	 Mouse	brain	Nuclei	N/A
8k Adult Mouse Cortex Cells, ATAC v1.1, Chromium X	Epi ATAC v1.1	 Mouse	brain	Nuclei	N/A
Multiomic Integration Neuroscience Application Note: Single Cell Multiome RNA + ATAC Alzheimer's Disease Mouse Model Brain Coronal Sections from One Hemisphere Over a Time Course	Epi Multiome ATAC + Gene Expression v1.0	 Mouse	brain	Nuclei	Fresh Frozen, FFPE
Mouse E18 Combined Cortex, Hippocampus and Subventricular Zone Nuclei Multiplexed, 12 CMOs: 3'v3.1 Targeted, Custom Neuroscience Panel	Universal 3' Gene Expression v3.1	 Mouse	brain, cortex, hippocampus, subventricular zone	Nuclei	N/A
Flash-Frozen Human Healthy Brain Tissue (3k)	Epi Multiome ATAC + Gene Expression v1.0	 Human	brain, cerebellum	Nuclei	Frozen
Fresh Embryonic E18 Mouse Brain (5k)	Epi Multiome ATAC + Gene Expression v1.0	 Mouse	cortex, hippocampus, ventricular zone, brain	Nuclei	Fresh

Dataset we will analyse today

- Alzheimer's disease model: ApoE knock out
- Single nucleus data
- 7 pooled hippocampi samples from 9-month-old C57BL/6 J wild type (WT) mice (ID GSM5067107)
- 7 pooled hippocampi samples from 9-month-old *ApoE* knockout (EKO) mice (ID GSM5067109)

differences in specific cell types.

All analyses were performed on GEO dataset [GSE166261](#) [dataset] (Shi et al., 2021a, b) comprised of snRNAseq data generated by droplet-based Chromium Single Cell 3' Reagent Kit (10x Genomics) on 7 pooled hippocampi samples from 9-month-old C57BL/6 J wild type (WT) mice (Sample GSM5067107) and 7 pooled hippocampi samples from 9-month-old *ApoE* knockout (EKO) mice (Sample GSM5067109). The initial EKO mouse was generated using the 129P2/OlaHsd-derived E14Tg2a ES cell line (Piedrahita et al., 1992). The mice used to generate the samples for the snRNAseq data for this study were produced by backcrossing at least 10 generations to C57BL/6 J inbred mice.



Short Communication

Analysis of differential gene expression and transcript usage in hippocampus of *ApoE* null mutant mice: Implications for Alzheimer's disease

Andrew E. Weller*, Glenn A. Doyle, Benjamin C. Reiner, Richard C. Crist, Wade H. Berrettini

Center for Neurobiology and Behavior, Department of Psychiatry, Perelman School of Medicine, University of Pennsylvania, Philadelphia, PA, 19104, United States

General workflow

- Data import and Seurat object generation (already done)
- QC for mitochondrial genes, nr of genes and expression level
- Normalization: normalizing gene expression level for each cell
- Feature selection of variable genes
- Scaling: scaling expression for each gene to avoid overrepresentation of highly expressed genes
- Linear dimensional reduction: PCA
- Clustering: k-means for cluster selection
- Non-linear dimensional reduction: UMAP
- Feature selection and cluster annotation: looking at cluster specific markers
- (optional) Cell mapping, subclustering

How to find the data (example: GEO Gene Expression Omnibus)

ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSM5067109

Scope: Self Format: HTML Amount: Quick GEO accession: GSE166261 GO

Series GSE166261 Query DataSets for GSE166261

Status Public on Feb 06, 2021

Title Overexpressing low-density lipoprotein receptor reduces tau-associated neurodegeneration via apoE-dependent and independent mechanisms

Organism [Mus musculus](#)

Experiment type Expression profiling by high throughput sequencing

Samples (6) [Less...](#)

- [GSM5067107](#) Isolated single nuclei from 7 pooled hippocampi of 9-month old WT mice
- [GSM5067108](#) Isolated single nuclei from 7 pooled hippocampi of 9-month old LDLR mice
- [GSM5067109](#) Isolated single nuclei from 7 pooled hippocampi of 9-month old EKO mice
- [GSM5067110](#) Isolated single nuclei from 8 pooled hippocampi of 9-month old P301S mice
- [GSM5067111](#) Isolated single nuclei from 7 pooled hippocampi of 9-month old P301S/LDLR mice
- [GSM5067112](#) Isolated single nuclei from 7 pooled hippocampi of 9-month old P301S/EKO mice

differences in specific cell types.

All analyses were performed on GEO dataset [GSE166261](#) [dataset] ([Shi et al., 2021a, b](#)) comprised of snRNAseq data generated by droplet-based Chromium Single Cell 3' Reagent Kit (10x Genomics) on 7 pooled hippocampi samples from 9-month-old C57BL/6 J wild type (WT) mice (Sample GSM5067107) and 7 pooled hippocampi samples from 9-month-old *ApoE* knockout (EKO) mice (Sample GSM5067109). The initial EKO mouse was generated using the 129P2/OlaHsd-derived E14Tg2a ES cell line ([Piedrahita et al., 1992](#)). The mice used to generate the samples for the snRNAseq data for this study were produced by backcrossing at least 10 generations to C57BL/6 J inbred mice.

Series (1) [GSE166261](#) Overexpressing low-density lipoprotein receptor reduces tau-associated neurodegeneration via apoE-dependent and independent mechanisms

Relations

BioSample [SAMN17812771](#)
SRA [SRX10038115](#)

Download all 3 files

Supplementary file	Size	Download	File type/resource
GSM5067109_EKO_barcode.tsv.gz	23.9 Kb	(ftp) (http)	TSV
GSM5067109_EKO_genes.tsv.gz	212.7 Kb	(ftp) (http)	TSV
GSM5067109_EKO_matrix.mtx.gz	14.7 Mb	(ftp) (http)	MTX

[SRA Run Selector](#) [?](#)

Raw data are available in SRA

Processed data provided as supplementary file

SEURAT

R toolkit for single cell genomics

Introduction into Seurat



satijalab.org/seurat/

Seurat 5.2.0

Install

Get started

Vignettes ▾

Extensions

FAQ

News

R

SEURAT

R toolkit

Introductory Vignettes

PBMC 3K guided tutorial

Data visualization vignette

SCTransform, v2 regularization

Using Seurat with multi-modal data

Seurat v5 Command Cheat Sheet

Data Integration

Introduction to scRNA-seq integration

<https://satijalab.org/seurat/>

Seurat v5

ata

Introduction into Seurat

- What it is
- Why we use it

SEURAT OBJECT

Cell ID	nFeature	nCount	Percent_mito	Percent_ribo	...	droplet_status	Cell_type
Cell_1	18500	1000	17	2		singlet	99
Cell_2	16070	700	12	0		singlet	95
Cell_3	17780	980	5	1		singlet	92
...	18000	1600	25	5		doublet	89
Cell_50000	17070	2400	7	10		singlet	100

CELL
METADATA

DIMENSIONALITY REDUCTION DATA

Gene ID	PCA_1	PCA_2	...	UMAP_1	UMAP_2
TP53	5	5		0.89	0.89
EGFR	3.5	3.5		1.2	1.2
VEGFA	0.78	0.78		2.2	2.2
...	3	3		0.76	0.76
BRCA1	1.2	1.2		2.4	2.4

GRAPH DATA

Cell ID	Cell_1	Cell_2	Cell_3	...
Cell_1	1	1	1	1
Cell_2	1	1	1	1
Cell_3	1	1	1	1
...	0	0	1	1
Cell_50000	1	1	0	0

COUNTS TABLE

Gene ID	Cell_1	Cell_2	Cell_3	Cell_4	Cell_5	Cell_6	...	Cell_50000
TP53	80	50	63	36	60	0		99
EGFR	9							
VEGFA	10							
APOE	95							
IL6	6							
TGFB1	82							
AKT1	7							
MTHFR	70							
...								
BRCA1	75							

Raw counts

Normalised counts

Scaled counts

GENE METADATA

Gene ID	Avg_expression	...	variance
TP53	5		0.89
EGFR	3.5		1.2
VEGFA	0.78		2.2
...	3		0.76
BRCA1	1.2		2.4

<https://biostatsquid.com/seurat-objects-explained/>

Introduction into Seurat

- What it is
- Why we use it

SEURAT OBJECT



EACH LAYER STORES A VERSION OF THE GENE EXPRESSION DATA

General workflow

- Data import and Seurat object generation (already done)
 - QC for mitochondrial genes, nr of genes and expression level
 - Normalization: normalizing gene expression level for each cell
 - Feature selection of variable genes
 - Scaling: scaling expression for each gene to avoid overrepresentation of highly expressed genes
 - Linear dimensional reduction: PCA
 - Clustering: k-means for cluster selection
 - Non-linear dimensional reduction: UMAP
-
- Feature selection and cluster annotation: looking at cluster specific markers
 - (optional) Cell mapping, subclustering

Part 1: Data import

```
library(dplyr)
library(Seurat)
library(patchwork)

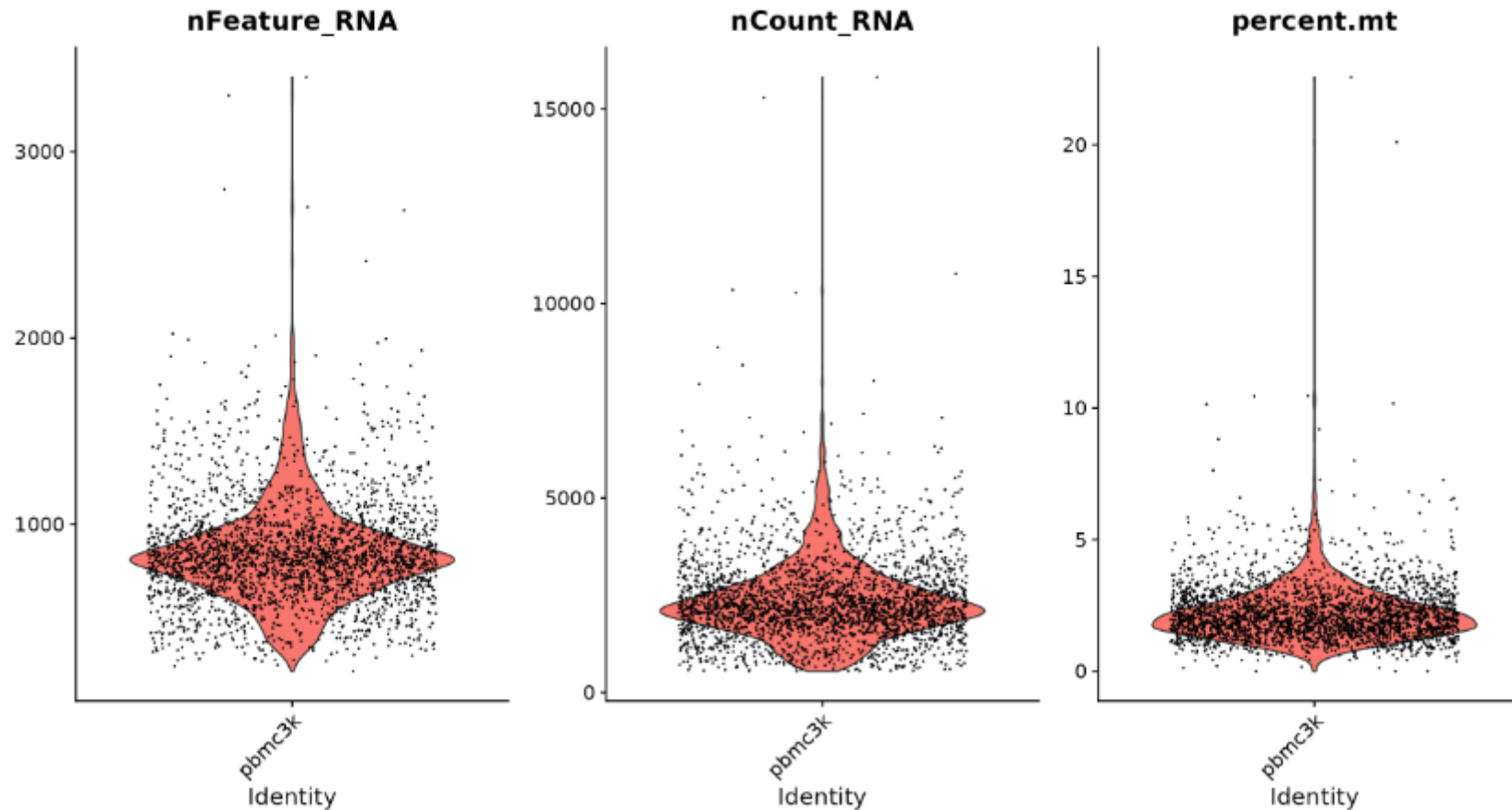
# Load the PBMC dataset
pbmc.data <- Read10X(data.dir = "/brahms/mollag/practice/filtered_gene_bc_matrices/hg19/")
# Initialize the Seurat object with the raw (non-normalized data).
pbmc <- CreateSeuratObject(counts = pbmc.data, project = "pbmc3k", min.cells = 3, min.features = 200)
pbmc
```

```
## An object of class Seurat
## 13714 features across 2700 samples within 1 assay
## Active assay: RNA (13714 features, 0 variable features)
## 1 layer present: counts
```

https://satijalab.org/seurat/articles/pbmc3k_tutorial

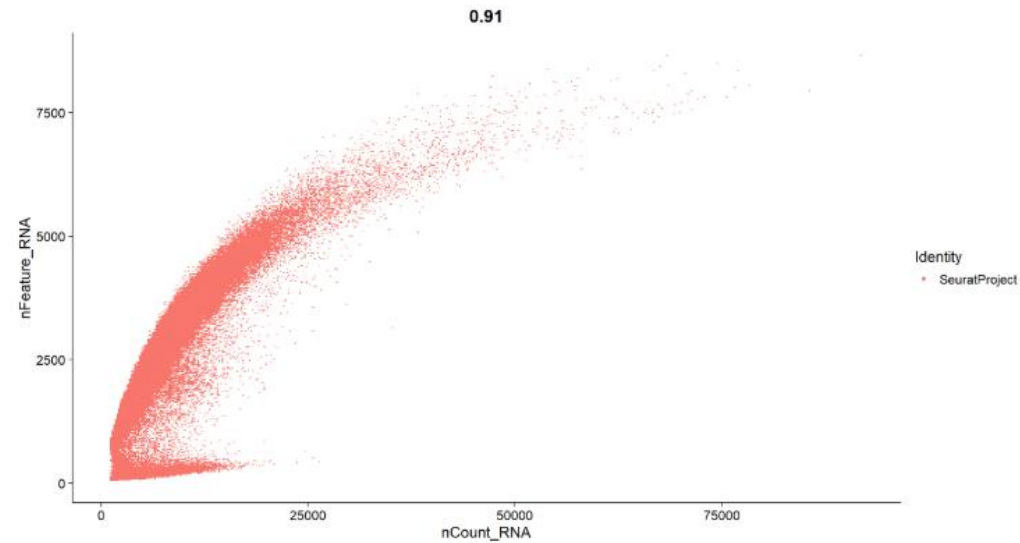
Part 2: Data filter

```
# Visualize QC metrics as a violin plot  
VlnPlot(pbmc, features = c("nFeature_RNA", "nCount_RNA", "percent.mt"), ncol = 3)
```

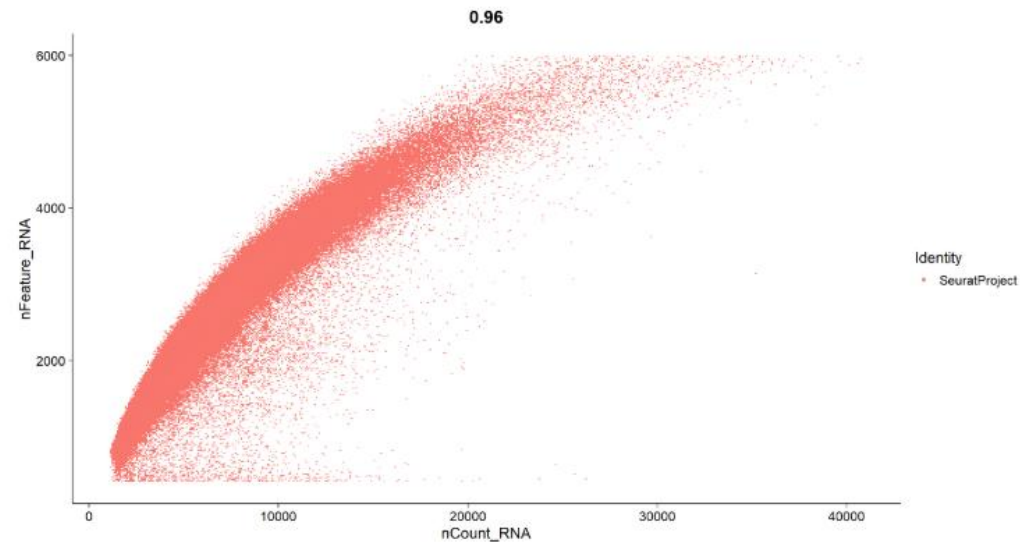


Part 2: Data filter

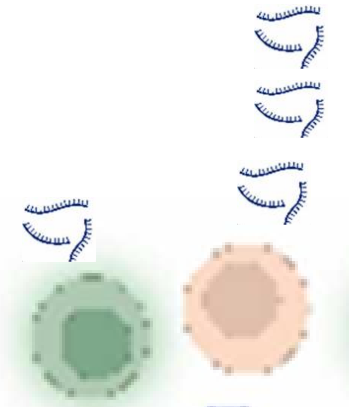
```
# FeatureScatter is typically used to visualize feature-feature relationships, but can be used for anything calculated by the object, i.e. columns in object metadata, PC scores etc.  
FeatureScatter(obj, feature1 = "nCount_RNA", feature2 = "nFeature_RNA")
```



```
obj <- subset(obj, subset = nFeature_RNA > 400 & nFeature_RNA < 6000)  
FeatureScatter(obj, feature1 = "nCount_RNA", feature2 = "nFeature_RNA")
```



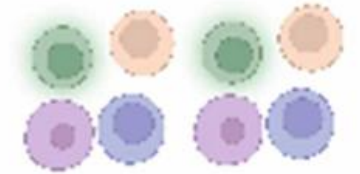
Normalization



	Cell 1	Cell 2	Cell 3
Total reads	3	300	100
RAW			
Gene A	1	100	50
Gene B	2	200	50
NORMALISED			
Gene A	8.11	8.11	8.52
Gene B	8.81	8.81	8.52

LOG2 NORMALISATION
↓

Gene A cell 1:
 $\log_2(1/3 * 10000 + 1)$



Normalization: This step addresses differences between cells by adjusting for sequencing depth or other technical variations, ensuring that gene expression values are comparable across cells.

Part 2: Data normalization

Normalizing the data

After removing unwanted cells from the dataset, the next step is to normalize the data. By default, we employ a global-scaling normalization method "LogNormalize" that normalizes the feature expression measurements for each cell by the total expression, multiplies this by a scale factor (10,000 by default), and log-transforms the result. In Seurat v5, Normalized values are stored in `obj[["RNA"]]` data.

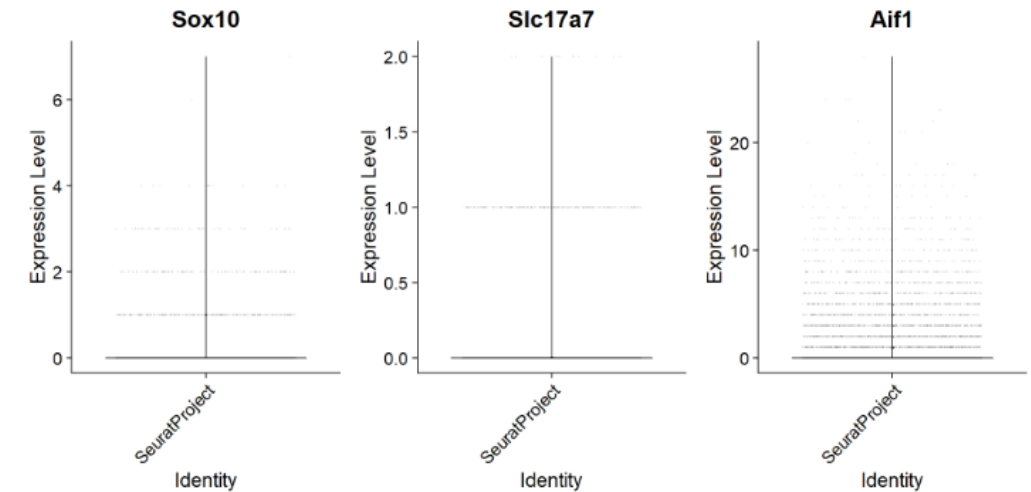
```
#obj <- NormalizeData(obj, normalization.method = "LogNormalize", scale.factor = 1e4)
```

For clarity, in this previous line of code (and in future commands), we provide the default values for certain parameters in the function call. However, this isn't required and the same behavior can be achieved with:

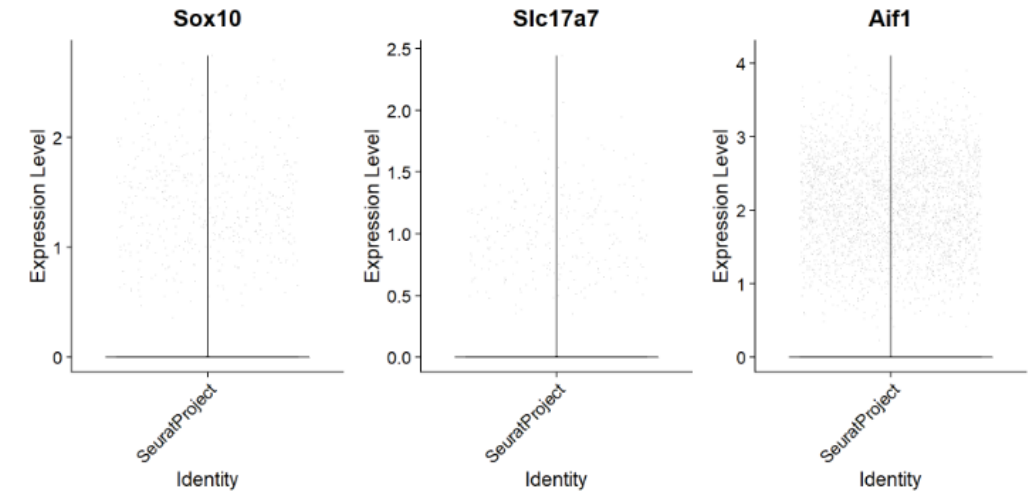
```
obj <- NormalizeData(obj)
```

Compare Normalization with Counts

```
VlnPlot(obj, features = c("Sox10", "Slc17a7", "Aif1"), ncol = 3, layer = "counts", alpha = 0.1)
```



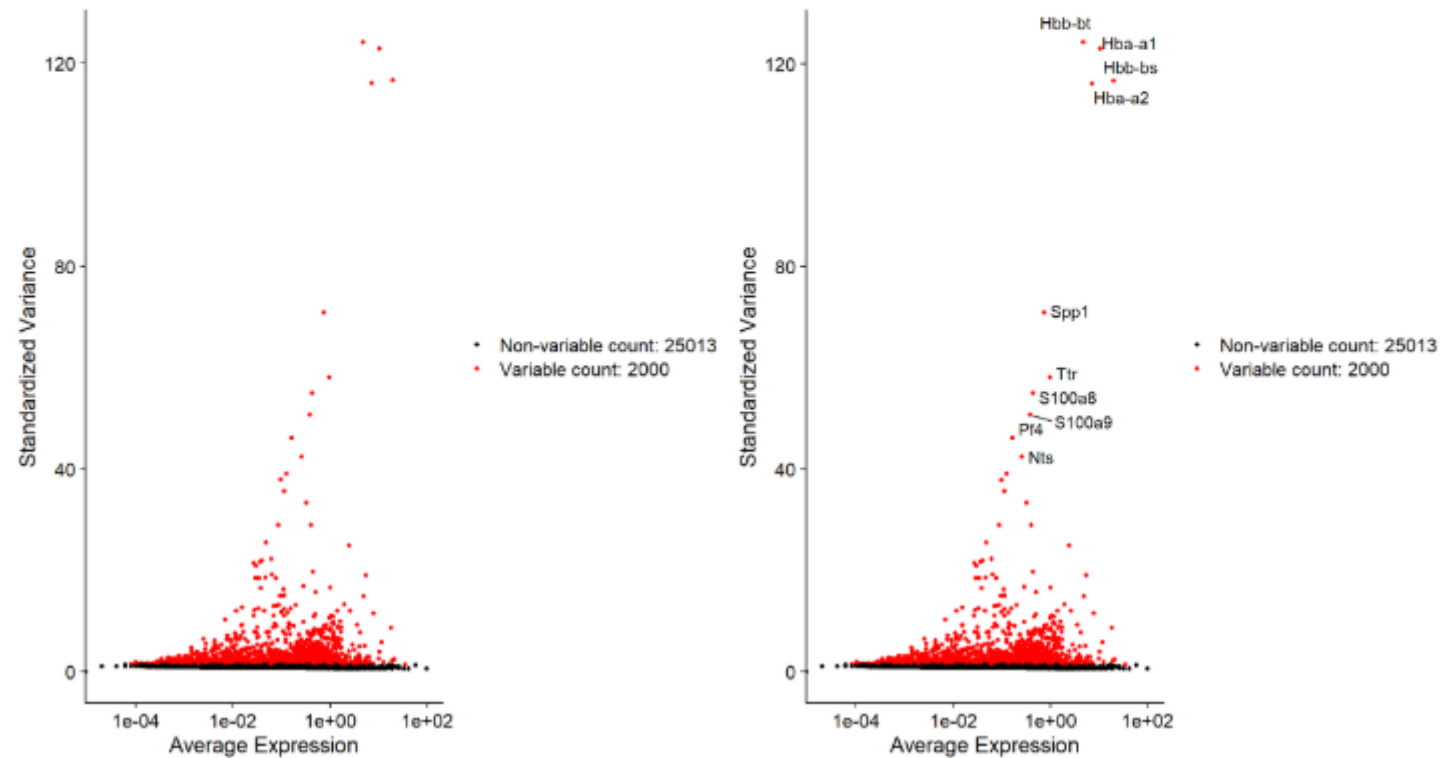
```
# We then visualize normalized data  
VlnPlot(obj, features = c("Sox10", "Slc17a7", "Aif1"), ncol = 3, layer = "data", alpha = 0.1)
```



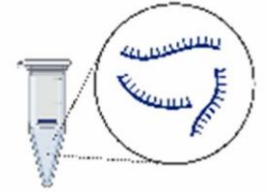
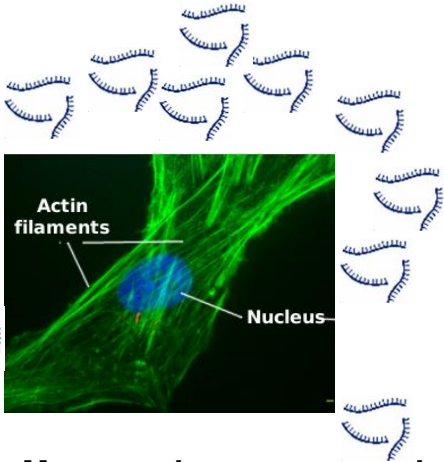
Part 3: Find variable features (feature selection)

```
obj <- FindVariableFeatures(obj)
# Identify the 10 most highly variable genes
top10 <- head(VariableFeatures(obj), 10)

# plot variable features with and without labels
plot1 <- VariableFeaturePlot(obj)
plot2 <- LabelPoints(plot = plot1, points = top10, repel = TRUE)
plot1 + plot2
```



Scaling



RAW	Cell 1	Cell 2	Cell 3
Gene A	1	100	50
Gene B	2	200	50

NORMALISED			
Gene A	8.11	8.11	8.52
Gene B	8.81	8.81	8.52

SCALED			
Gene A	-0.57	-0.57	+1.19
Gene B	+0.58	+0.58	-1.16

LOG2 NORMALISATION

Z-SCORE

Scaling: This step addresses differences between genes by adjusting for varying ranges in expression levels across genes, making sure that no single gene with very high expression dominates the analysis.

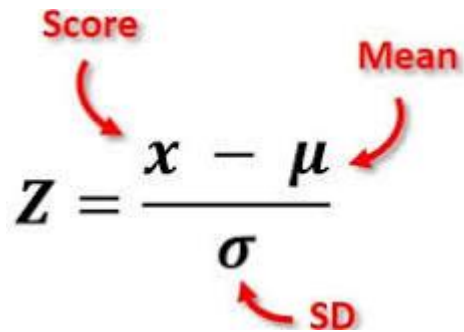
Part 4: scale the data

$$Z = \frac{x - \mu}{\sigma}$$

Score

Mean

SD



Scaling the data

Next, we apply a linear transformation ("scaling") that is a standard pre-processing step prior to dimensional reduction techniques like PCA. The `ScaleData()` function:

- Shifts the expression of each gene, so that the mean expression across cells is 0
- Scales the expression of each gene, so that the variance across cells is 1
 - This step gives equal weight in downstream analyses, so that highly-expressed genes do not dominate
- The results of this are stored in `obj[["sketch"]][scale.data]`
- By default, only variable features are scaled.
- You can specify the `features` argument to scale additional features

```
obj <- ScaleData(obj)
```

Part 5: PCA

For the first principal components, Seurat outputs a list of genes with the most positive and negative loadings, representing modules of genes that exhibit either correlation (or anti-correlation) across single-cells in the dataset.

```
obj <- RunPCA(obj, features = VariableFeatures(object = obj))
```

Seurat provides several useful ways of visualizing both cells and features that define the PCA, including `VizDimReduction()`, `DimPlot()`, and `DimHeatmap()`

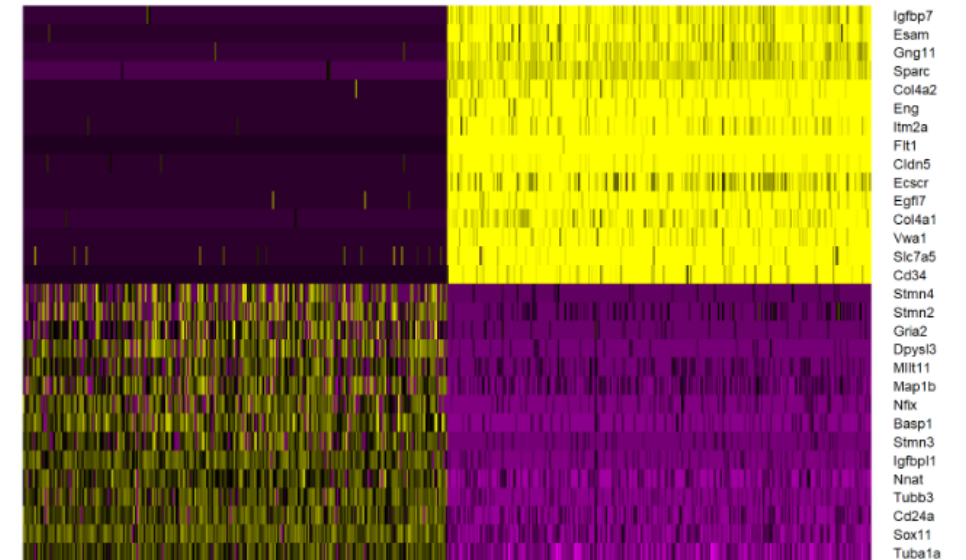
```
# Examine and visualize PCA results a few different ways
print(obj[['pca']], dims = 1:5, nfeatures = 5)
```

```
## PC_1
## Positive: Tuba1a, Sox11, Cd24a, Tubb3, Nnat
## Negative: Igfbp7, Esam, Gng11, Sparc, Col4a2
## PC_2
## Positive: Clqc, Clqb, Tyrobp, Clqa, Fcer1g
## Negative: Tmsb10, Tuba1a, Tsc22d1, Sparcl1, Serpinh1
## PC_3
## Positive: Tubb3, Tmsb10, Mllt11, Stmn2, Stmn3
## Negative: Hmgb2, Dbi, Phgdh, Fabp7, Cks2
## PC_4
## Positive: Vtn, Ndufa4l2, Kcnj8, Higd1b, Abcc9
## Negative: Cldn5, Vwa1, Cd34, Pglyrp1, Ctla2a
## PC_5
## Positive: Birc5, Ube2c, Ccna2, Nusap1, Spc25
## Negative: Cimap3, Clu, Foxj1, Pierce1, Rsph1
```

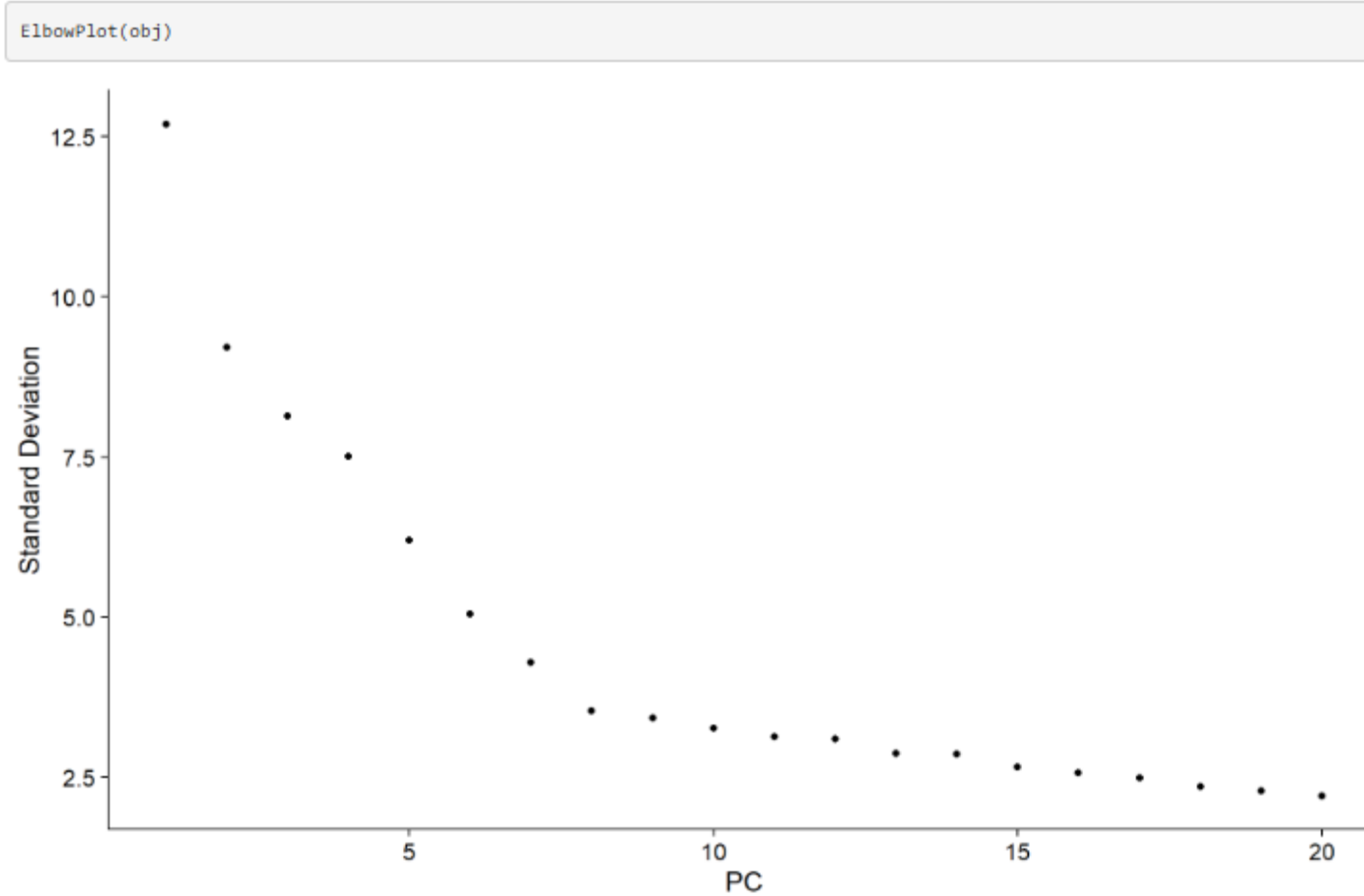
```
VizDimLoadings(obj, dims = 1:2, reduction = 'pca')
```

```
DimHeatmap(obj, dims = 1, cells = 500, balanced = TRUE)
```

PC_1



Part 5: PCA



Part 6: Clustering

```
obj <- FindNeighbors(obj,reduction = "pca", dims = 1:50)
obj <- FindClusters(obj, resolution = 2) # the higher the number the higher the clusters
```

```
## Modularity Optimizer version 1.3.0 by Ludo Waltman and Nees Jan van Eck
##
## Number of nodes: 49451
## Number of edges: 1969041
##
## Running Louvain algorithm...
## Maximum modularity in 10 random starts: 0.8779
## Number of communities: 54
## Elapsed time: 13 seconds
```

```
# Look at cluster IDs of the first 5 cells
head(Ids(obj), 5)
```

```
## AAACCTGAGATAGGAG-1 AAACCTGAGCGGCTTC-1 AAACCTGAGGAATCGC-1 AAACCTGAGGACACCA-1
## <NA> <NA> <NA> <NA>
## AAACCTGAGGCCCGTT-1
## 17
## 54 Levels: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 ... 53
```

Part 7: Non-linear dimensional reduction (UMAP)

Only WT show!!!

Day 4

Clustering of cells in Seurat

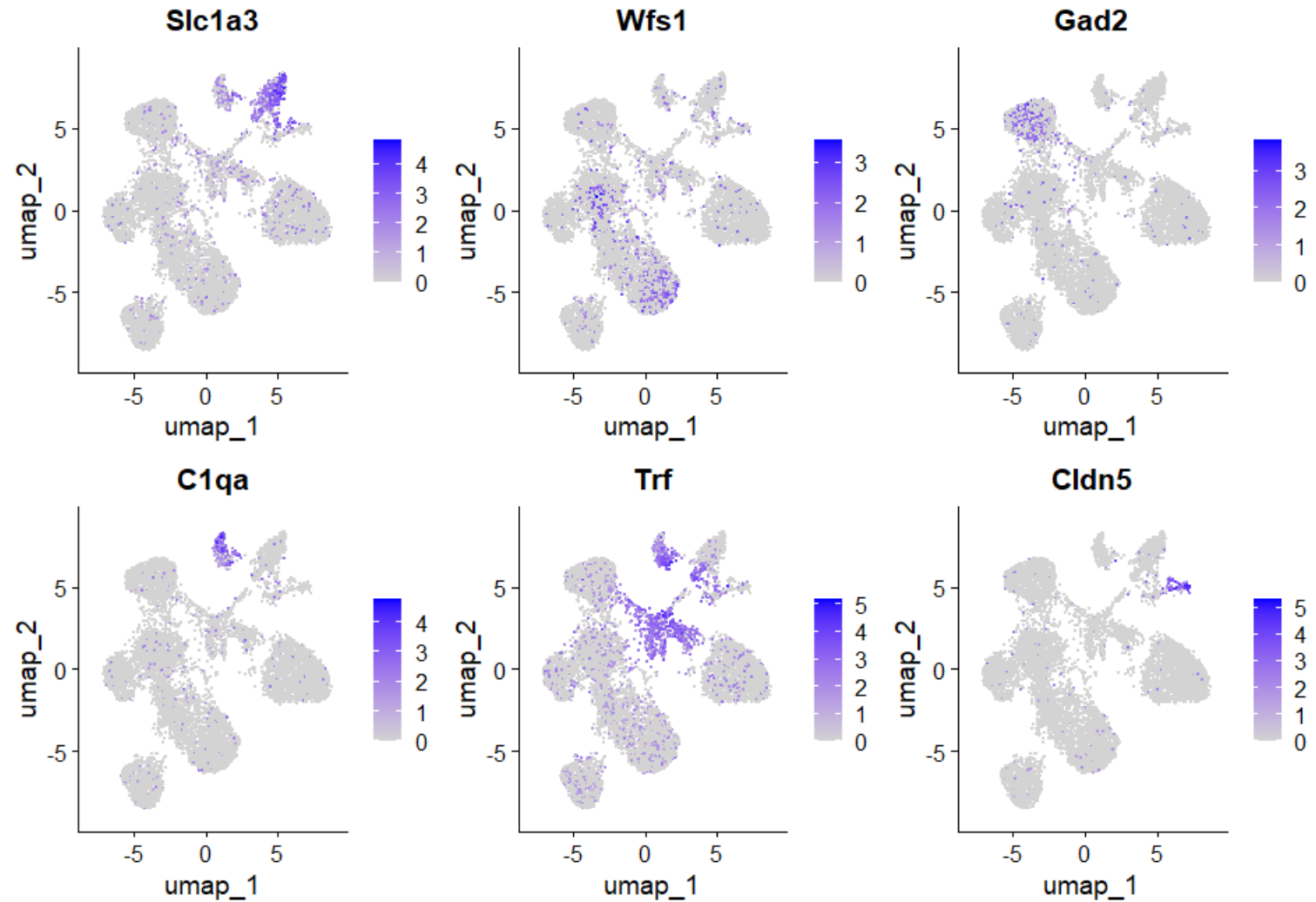
DEG and its interpretation

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Part 8: Find markers

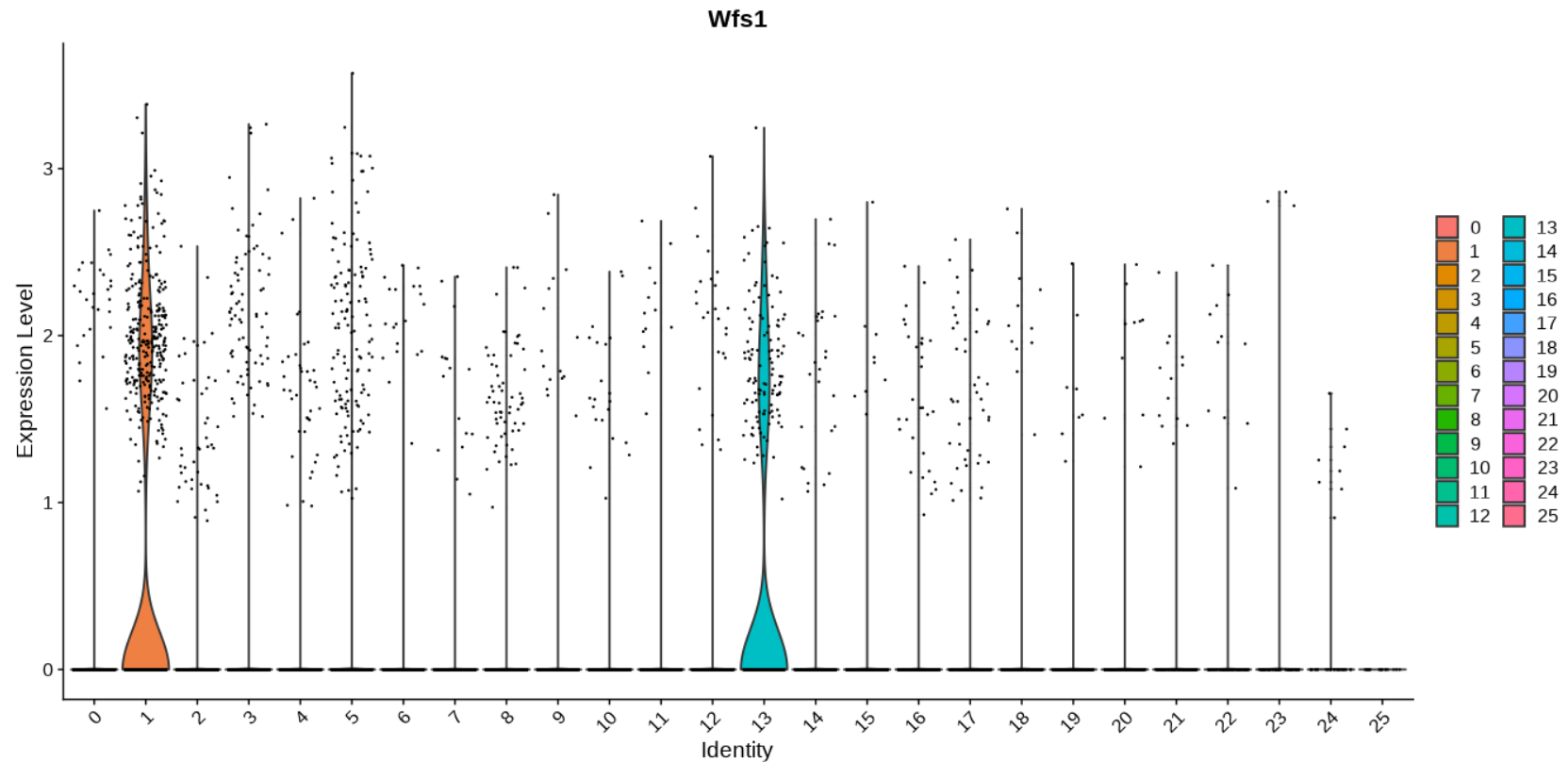
```
```{r,fig.height = 7, fig.width = 10}
FeaturePlot(
 object = Seurat_AD_WT,
 features = c(
 "Slc1a3", #Astrocytes
 "Wfs1", # Glut Neurons or Slc17a7
 "Gad2", # GABA neurons
 "C1qa", # Microglia
 "Trf", # Myelin-forming mature oligodendrocytes
 "Cldn5"), # Endothelial cells
 ncol = 3)
```
```



Part 8: Find markers

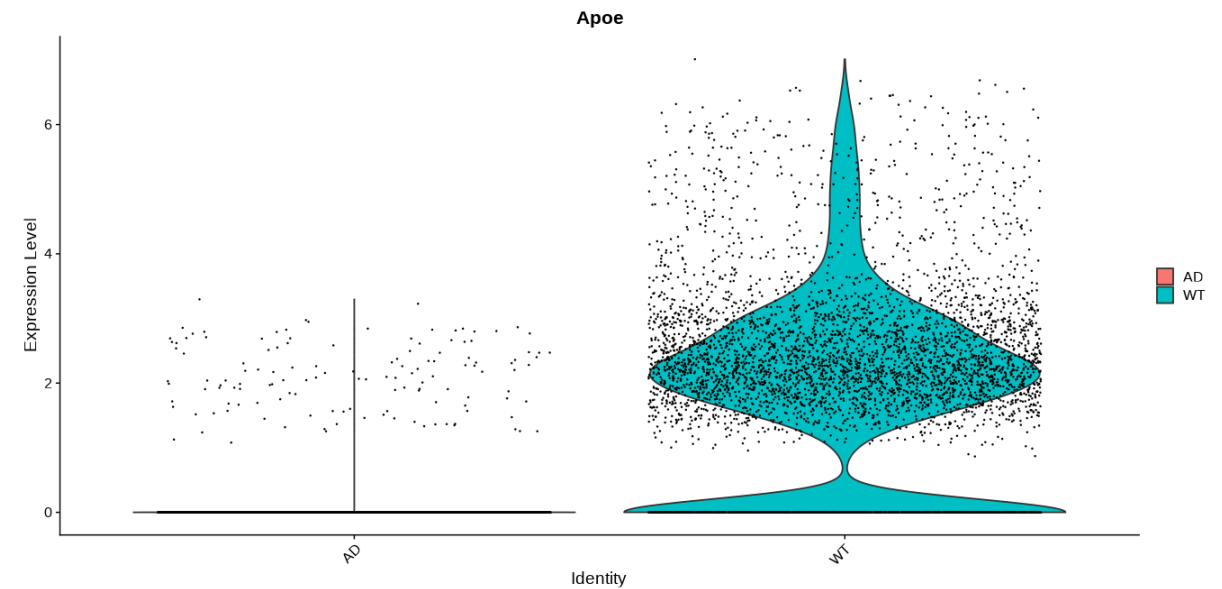
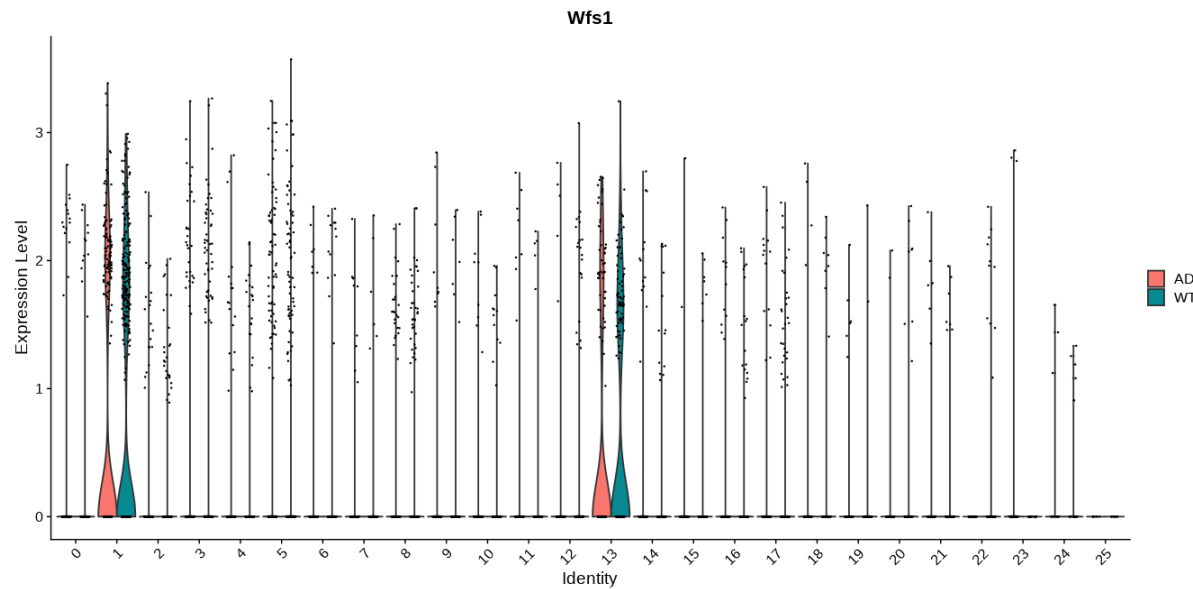
Differential gene expression

```
VlnPlot(obj, 'D1x2')
```



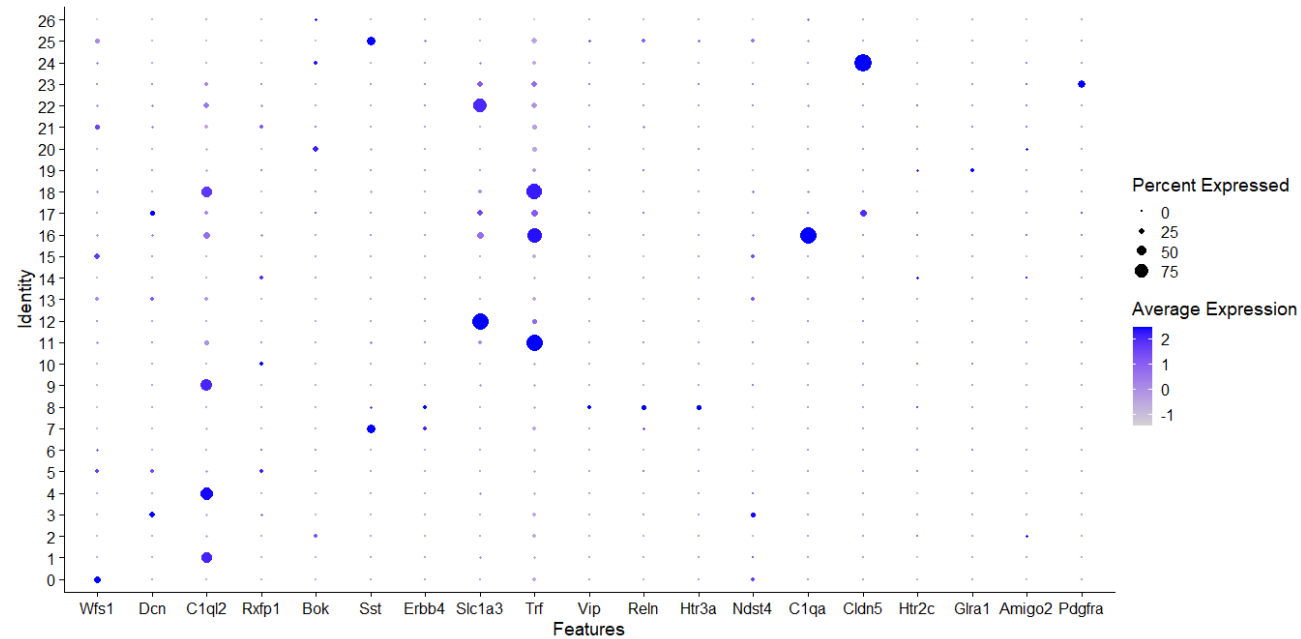
Part 8: Find markers

```
```{r, fig.height = 7, fig.width = 14}
VlnPlot(Seurat_AD_WT, 'Wfs1')
VlnPlot(Seurat_AD_WT, 'Wfs1', split.by = 'condition')
VlnPlot(Seurat_AD_WT, 'Apoe', group.by = 'condition') # check knockout
```
```



Part 8: Find markers

```
```{r, fig.height = 7, fig.width = 14}  
DotPlot(Seurat_AD_WT, features= all_genes)
```



# Part 8: Find markers

```
```{r, fig.height = 7, fig.width = 14}
# find all markers of cluster 5
Idents(Seurat_AD_WT) <- "seurat_clusters"
cluster13.markers <- FindMarkers(Seurat_AD_WT, ident.1 = "13", only.pos = TRUE)
head(cluster13.markers, n = 10)
```

Markers

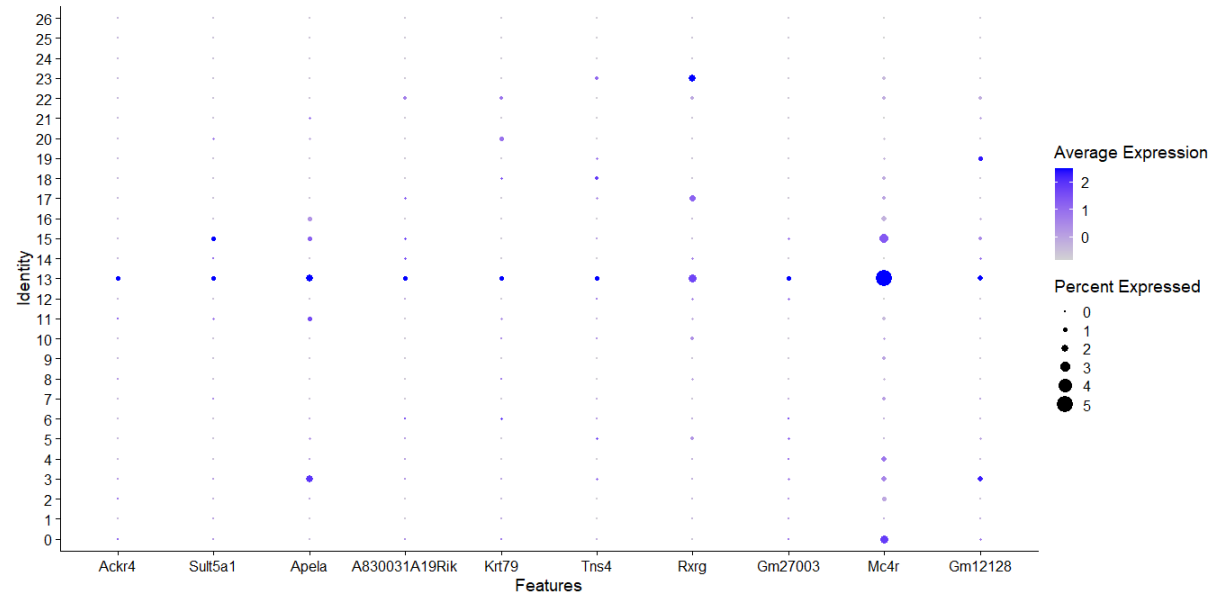
	p_val	avg_log2FC	pct.1	pct.2	p_val_adj
Cpne7	3.795059e-113	1.6148848	0.939	0.440	1.062541e-108
Pex5l	6.549430e-80	1.3841500	0.867	0.414	1.833709e-75
Man1a	1.168222e-55	1.7415038	0.408	0.123	3.270787e-51
Grin2b	6.088627e-55	0.6456170	1.000	0.913	1.704694e-50
Kcnn2	1.985708e-46	1.3211142	0.589	0.252	5.559586e-42
3110035E14Rik	1.610460e-42	1.1141428	0.614	0.279	4.508965e-38
Ntn	1.879055e-40	0.7167477	0.956	0.618	5.260977e-36
Chrd	8.107074e-40	1.0820214	0.683	0.354	2.269819e-35
Fam19a1	3.278474e-37	0.6712845	0.917	0.568	9.179072e-33
Fibcd1	4.322958e-37	1.6778961	0.267	0.076	1.210342e-32
Zeb2	6.237067e-33	0.7754559	0.853	0.579	1.746254e-28
Arhgef28	8.172101e-31	1.3445675	0.294	0.099	2.288025e-26
Mpped1	4.221407e-29	1.0897637	0.481	0.220	1.181910e-24
Brd9	7.888922e-28	0.7317513	0.836	0.619	2.208740e-23
Cadps2	6.051534e-27	1.3831535	0.256	0.086	1.694308e-22
Atp2b1	1.206030e-26	0.5222782	0.939	0.733	3.376643e-22
Grm5	2.330329e-26	0.5810710	0.939	0.721	6.524456e-22

- **avg_logFC**: log fold-change of the average expression between the two groups. Positive values indicate that the gene is more highly expressed in the first group
- **pct.1**: The percentage of cells where the gene is detected in the first group
- **pct.2**: The percentage of cells where the gene is detected in the second group
- **p_val_adj**: Adjusted p-value, based on bonferroni correction using all genes in the dataset

Part 8: Find markers

```
### Plot the markers
```{r, fig.height = 7, fig.width = 14}
cluster13.markers %>%
 dplyr::filter(avg_log2FC > 2 & p_val < 0.05) %>%
 slice_max(avg_log2FC, n = 10) -> top10

Idents(Seurat_AD_WT) <- "seurat_clusters"
DotPlot(Seurat_AD_WT, features = rownames(top10))
DotPlot(Seurat_AD_WT, features = rownames(top10), group.by = "condition")
```
```



Part 8: Find markers

| Aspect | FindMarkers(ident.1 = 5) | FindMarkers(ident.1 = "5_WT",
ident.2 = "5_KO") | FindConservedMarkers(ident.1 = 5,
grouping.var = "condition") |
|---------------------|---|---|---|
| Purpose | Find genes that define cluster 5 identity | Find genes differentially expressed between conditions in cluster 5 | Find genes that consistently mark cluster 5 regardless of condition |
| Comparison | Cluster 5 vs. all other cells | WT cluster 5 vs. KO cluster 5 | Within-condition cluster-vs-cluster comparisons, then combined |
| Output significance | Single p-value per gene | Single p-value per gene | Separate p-values per condition + meta-analysis p-value |
| Biological question | What makes cluster 5 unique? | Which genes respond differently to genotype in cluster 5? | Which genes robustly identify cluster 5 across both genotypes? |

MapMyCells

```
# Export data to MapMyCells
We will export the count matrix and map the cells with Allen Brain atlas
see: https://portal.brain-map.org/atlas-and-data/bkp/mapmycells
use: https://knowledge.brain-map.org/mapmycells/process/
## Extract the Count Matrix

```{r,fig.height = 7, fig.width = 10}
mat <- GetAssayData(object = subset(Seurat_AD_WT, downsample = 1000, subset = seurat_clusters %in% c(13)), assay = "RNA", slot = "counts")
we need to transpose the matrix, as MapMyCells expects matrix in which rows are "cells" and columns are genes!
mat_t <- t(mat)
write.csv(mat_t, "./output/count_matrix_cluster13.csv")
```
```

MapMyCells

Step 1

Upload your gene expression data
Input file requirements, limits, and creation.


[Click to upload](#) or drag and drop
anndata or csv (max. 2GB)

 CITE THIS TOOL

☐ Notify me when my mapping concludes

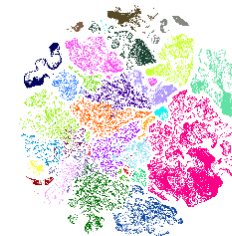
 Want to save your data? [SIGN IN](#)

Data Usage & Privacy

Allen Institute does not use, retain, or aggregate any data uploaded to MapMyCells for its own internal purposes, nor will we publish your data publicly. Allen Institute database administrators can access any uploaded dataset for debugging and other error remediation purposes. All files will be deleted one week after upload. Please do not submit any sensitive data, personally identifiable data, or protected health data that could put an individual's privacy at risk into MapMyCells. See the Allen Institute Privacy Policy for more information on our privacy practices.

Step 2

Choose a reference taxonomy and mapping algorithm
Learn about available cell type references, algorithms, and output files.



Reference Taxonomy

10x Whole Mouse Brain (CCN20230722)

Mapping Algorithm

Hierarchical Mapping

 START

Thank you for the attention

Please evaluate our course!

Lukasz

Kaja